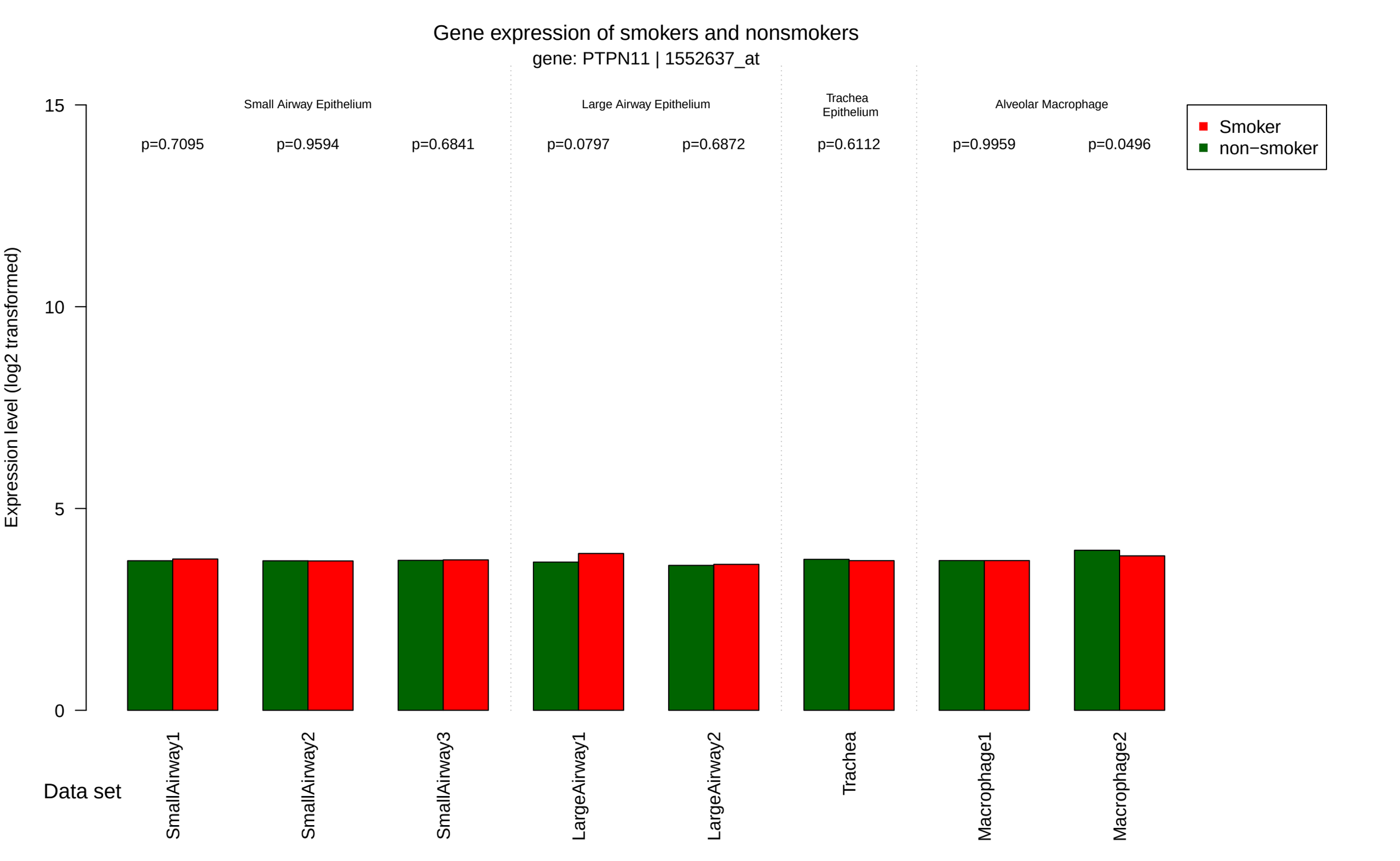
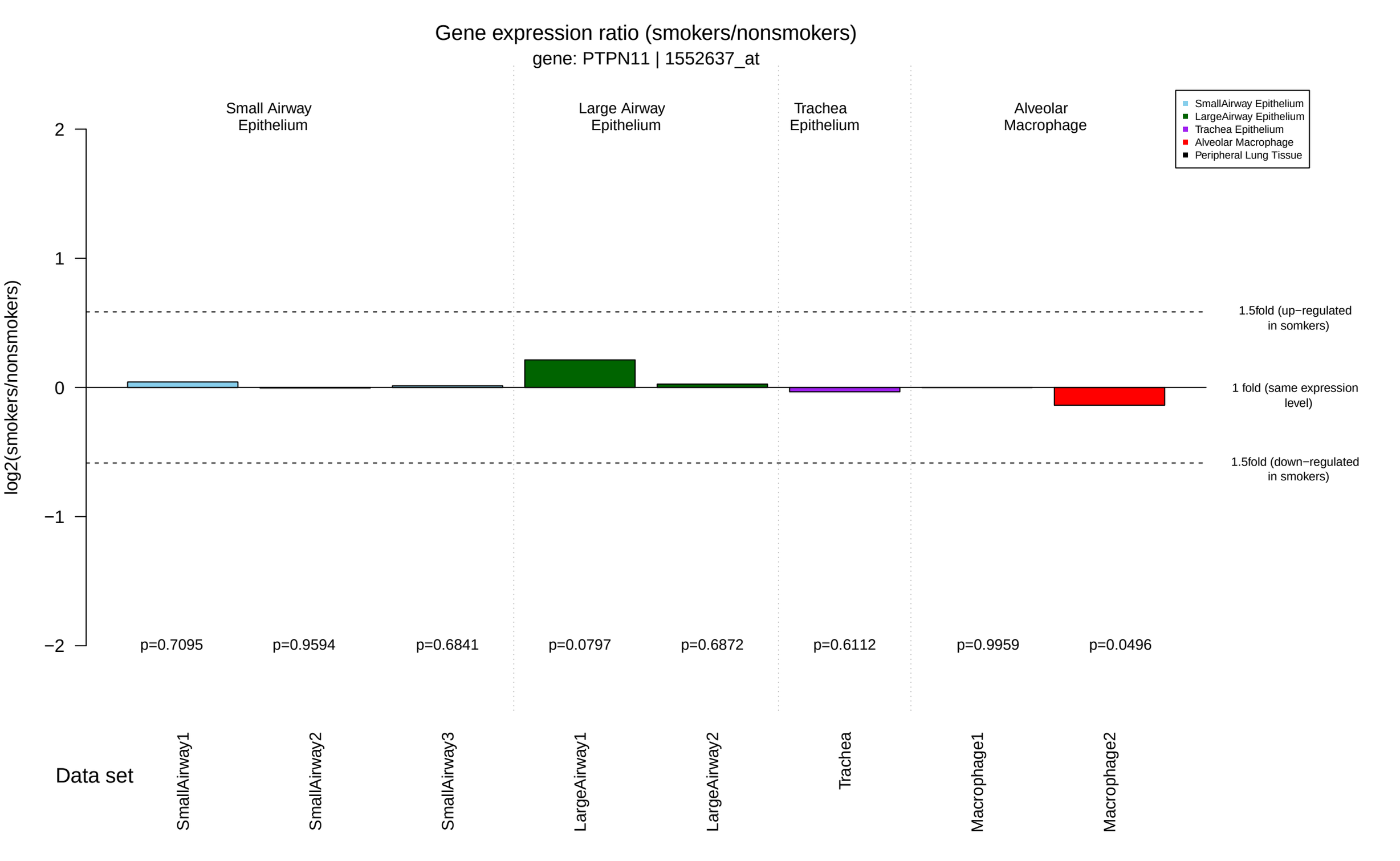
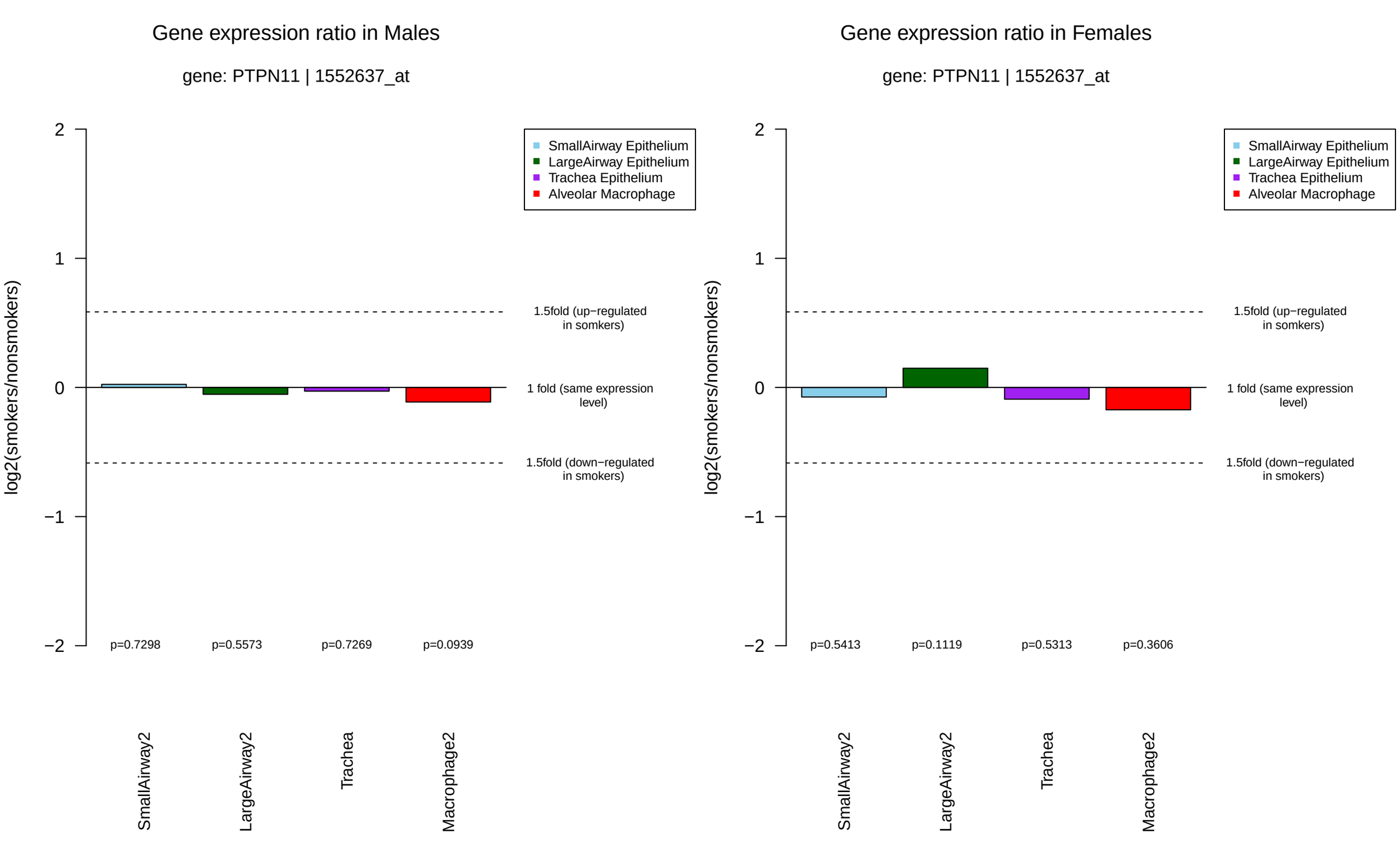
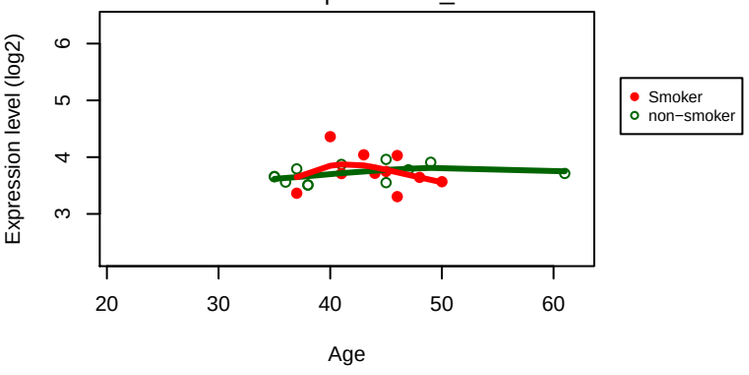
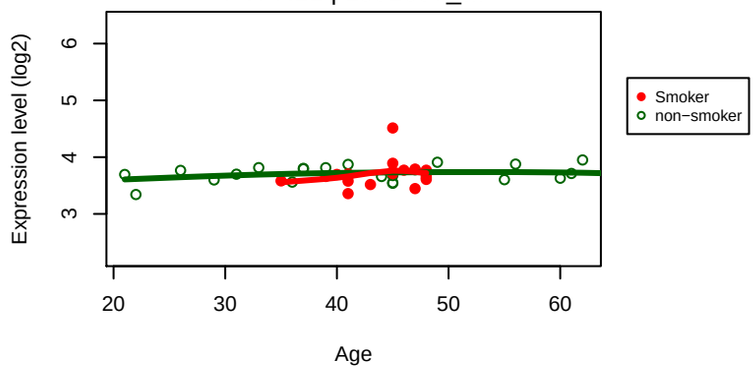
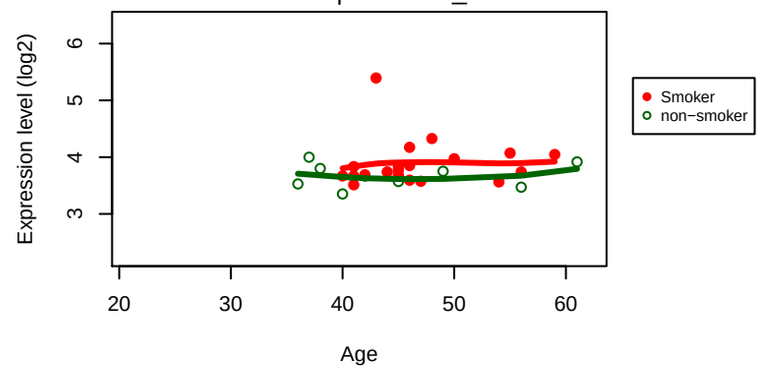

Gene expression of smokers and nonsmokers
gene: PTPN11 | 1552637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7095
p=0.9594
p=0.6841
p=0.0797
p=0.6872
p=0.6112
p=0.9959
p=0.0496
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

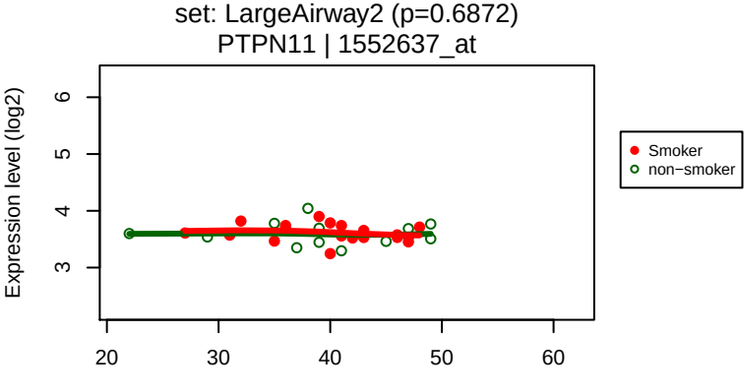

Gene expression ratio (smokers/nonsmokers)
gene: PTPN11 | 1552637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7095
p=0.9594
p=0.6841
p=0.0797
p=0.6872
p=0.6112
p=0.9959
p=0.0496
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

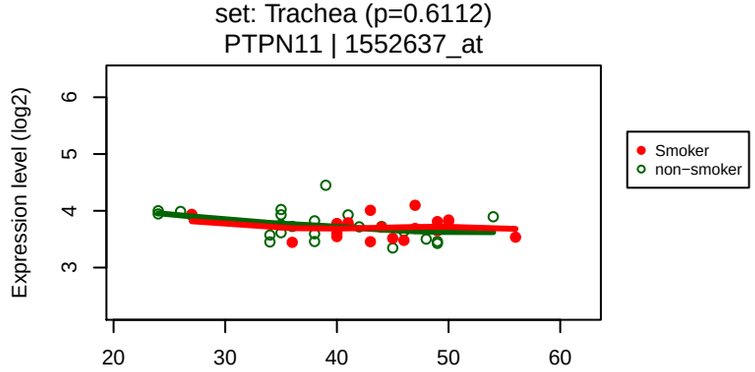

Gene expression ratio in Males
gene: PTPN11 | 1552637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7298
p=0.5573
p=0.7269
p=0.0939
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN11 | 1552637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5413
p=0.1119
p=0.5313
p=0.3606
SmallAirway2
LargeAirway2
Trachea
Macrophage2

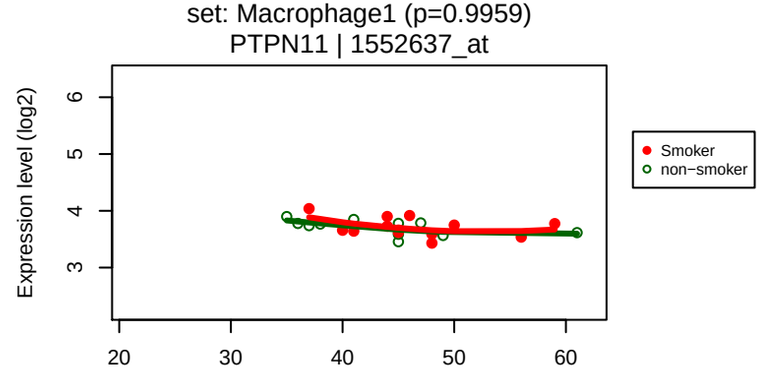

set: SmallAirway1 (p=0.7095)
PTPN11 | 1552637_at
Expression level (log2)
Age
Smoker
non−smoker

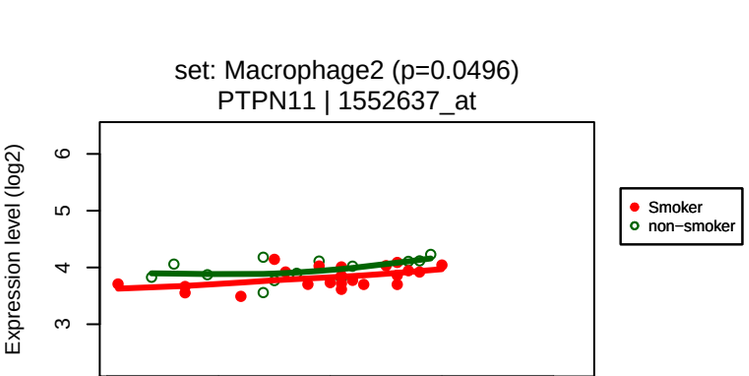

set: SmallAirway2 (p=0.9594)
PTPN11 | 1552637_at
Expression level (log2)
Age
Smoker
non−smoker

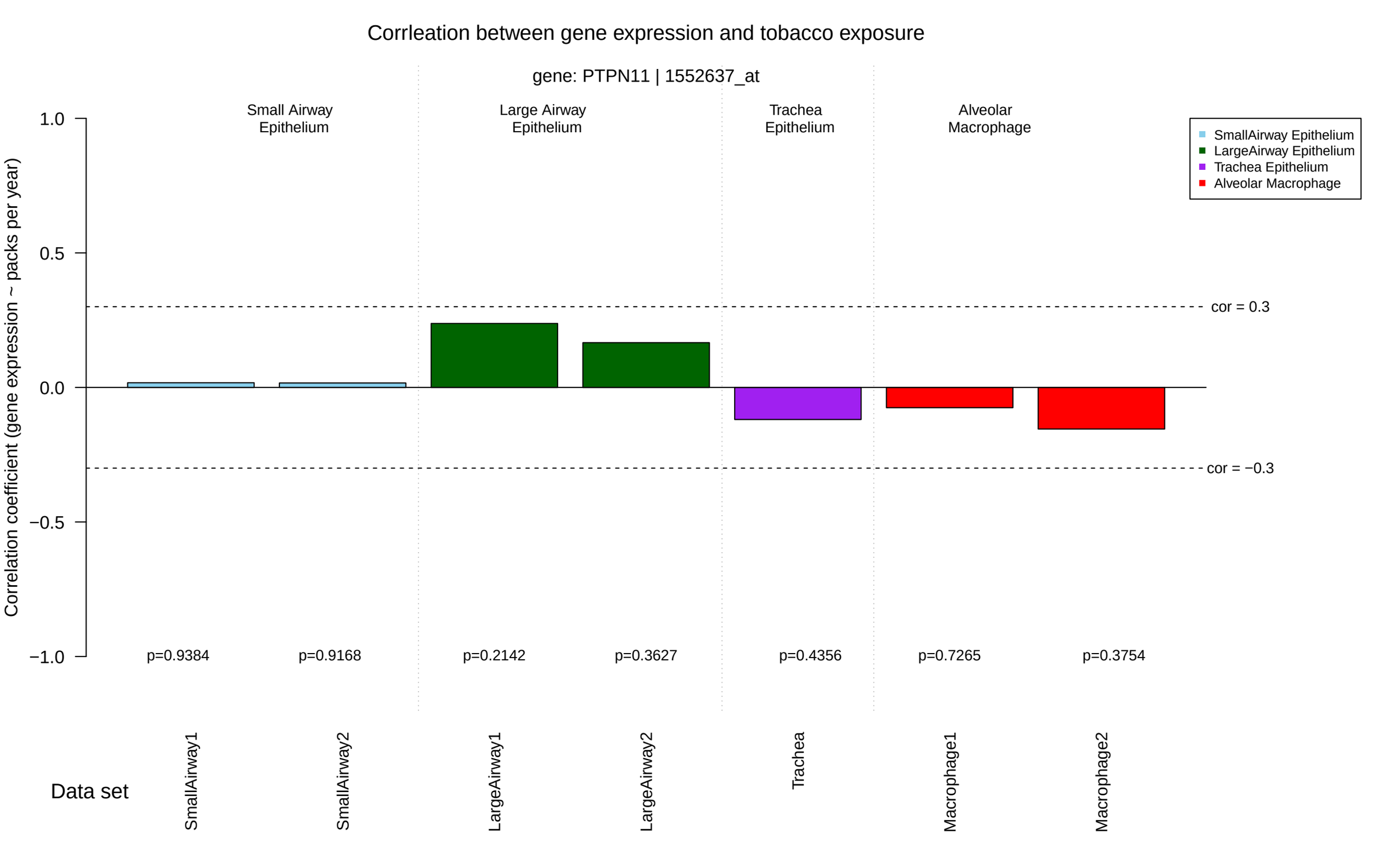

set: LargeAirway1 (p=0.0797)
PTPN11 | 1552637_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6872)
PTPN11 | 1552637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6112)
PTPN11 | 1552637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9959)
PTPN11 | 1552637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0496)
PTPN11 | 1552637_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN11 | 1552637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9384
p=0.9168
p=0.2142
p=0.3627
p=0.4356
p=0.7265
p=0.3754
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2